

MyUniPath

Rule-Based ICT Career Decision Support System

Technical Documentation

Tech Stack • Context Diagram • DFD (Level 1 & 2) • ERD • UML

Final Year Project | College of Computing & Informatics, UNITEN

1. Tech Stack

MyUniPath is built as a decoupled web application. The React frontend talks to the Laravel API exclusively via asynchronous JSON requests, and the API persists data in MySQL.

Layer	Technology	Purpose
Frontend Framework	React 18 + TypeScript	Component-driven SPA
Build Tool	Vite 6	Dev server + bundler
Styling	Tailwind CSS v4	Utility-first CSS
UI Primitives	shadcn/ui + Radix UI	Accessible components
Animation	Framer Motion	Page & UI transitions
Routing	React Router v6	Client-side routes
Charts	Recharts	Radar, pie, bar charts
State	React Context (StudentContext)	Auth + student session
Icons	lucide-react	Iconography
Backend Framework	Laravel 11	REST API & MVC
Language	PHP 8.3	Runtime
ORM	Eloquent	DB abstraction
Database	MySQL 8	Relational store
Auth	Custom session-id (X-Session-Id header)	Stateless API auth
Admin Auth	Admin password (X-Admin-Password header)	Admin endpoints
Asset Hosting	Laravel public/storage (transcripts)	PDF uploads
HTTP Client	Fetch API (typed wrapper in lib/api.ts)	Frontend ↔ Laravel
Package Mgmt	npm + Composer	JS & PHP dependencies
Dev Server	Vite (port 3000) + php artisan serve (8000)	(8000) dev

Architecture in one line

Browser ■ React (Vite, :3000) ■ HTTP JSON ■ Laravel API (:8000) ■ Eloquent ORM ■ MySQL

2. Context Diagram (DFD Level 0)

The whole system is shown as a single process. External entities represent the actors that exchange data with MyUniPath.

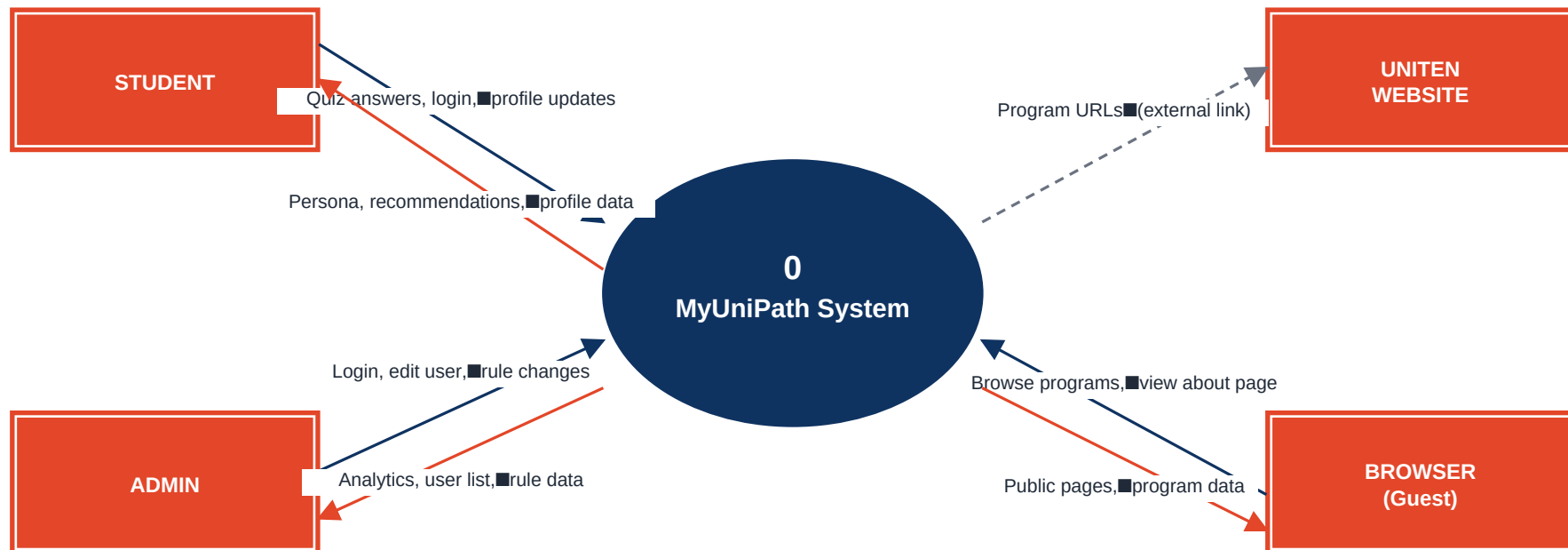


Figure 1 — Context Diagram. Inputs in navy, outputs in orange; dashed lines represent external (uniten.edu.my) links opened in a new tab.

3. Data Flow Diagram (Level 1)

Decomposition of the single process from the context diagram into the five main sub-processes. Data stores (D1–D6) map 1:1 onto MySQL tables.

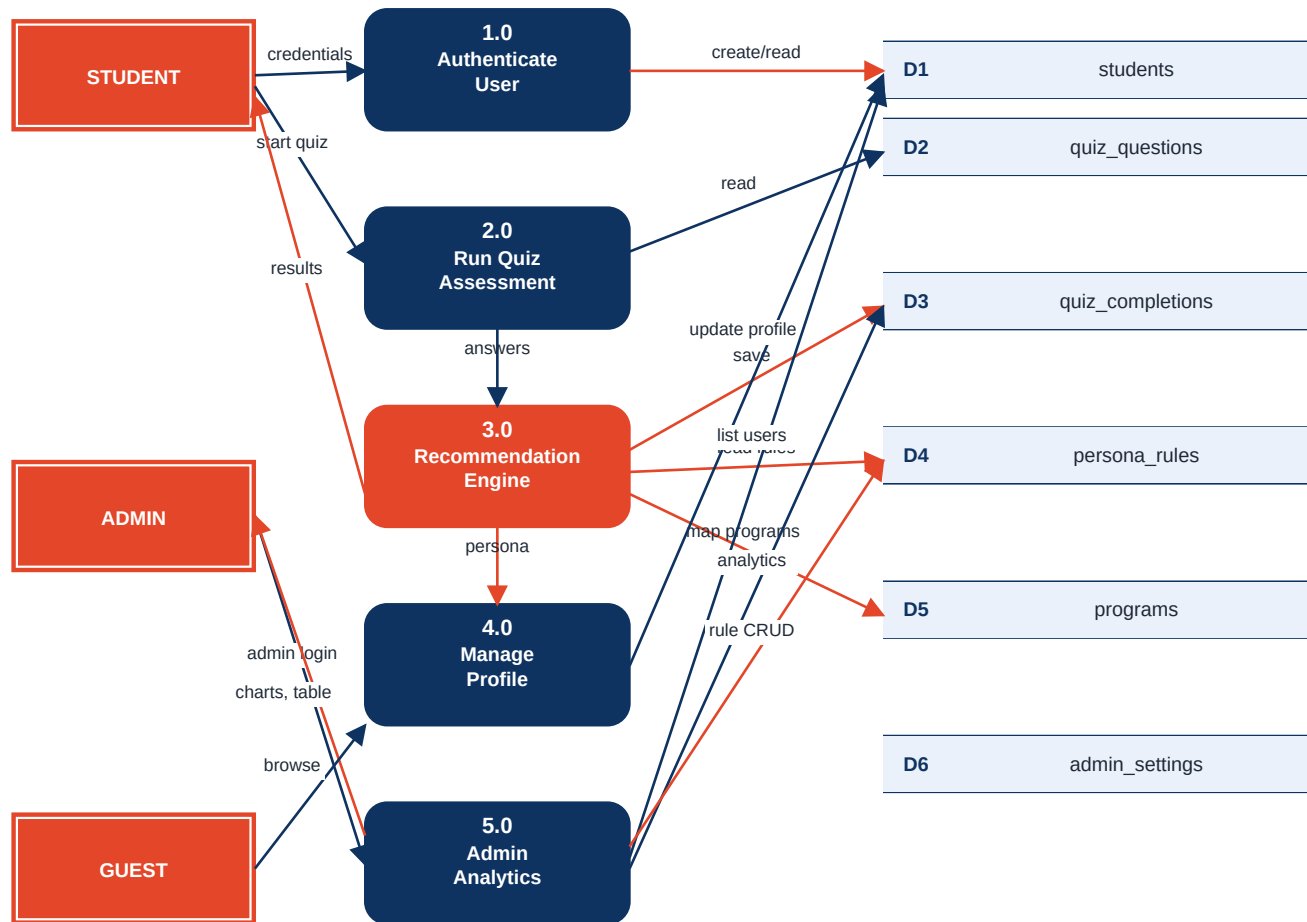


Figure 2 — Level 1 DFD. Orange flows indicate outputs back to the user / writes to data stores; navy flows indicate inputs / reads.

4. DFD Level 2 — Recommendation Engine (3.0)

Further decomposition of process 3.0 from the Level 1 DFD. Shows exactly how the engine turns a list of selected answer texts into a persona and a list of recommended program IDs.

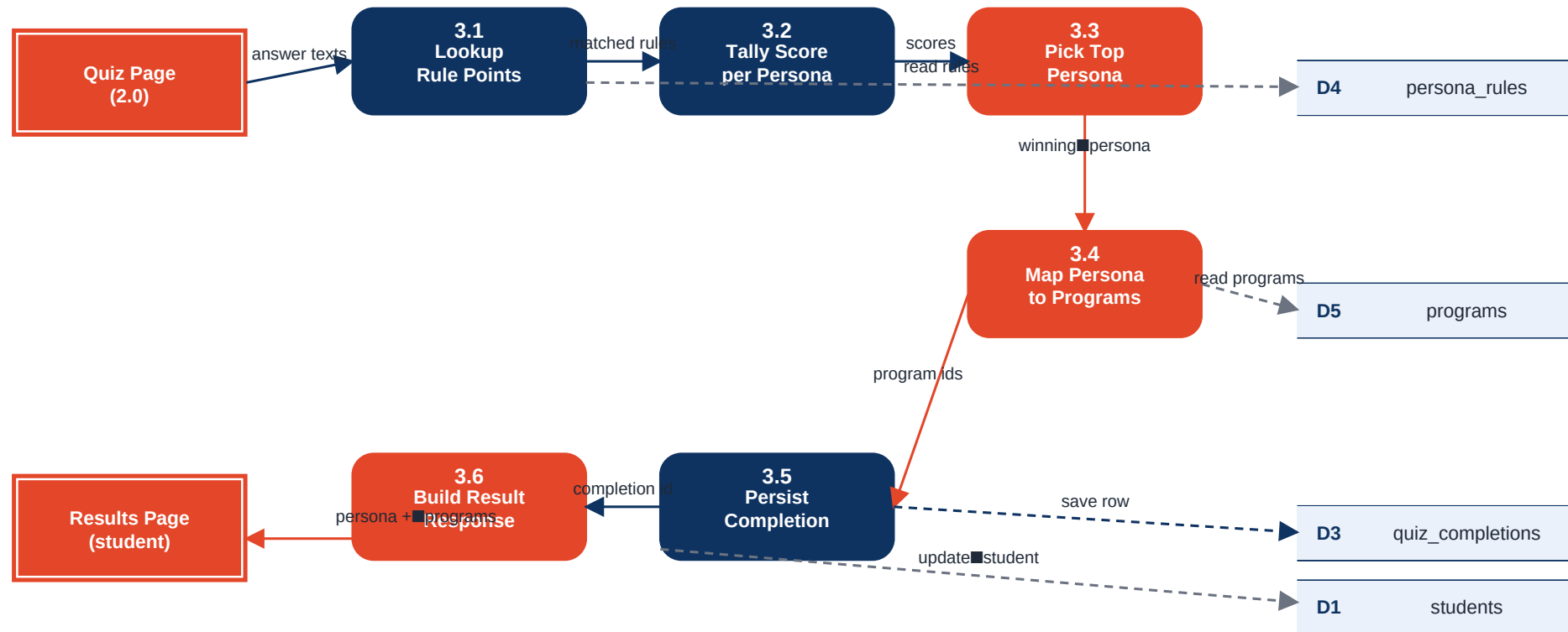


Figure 3 — Level 2 DFD for the Recommendation Engine. Dashed lines are reads / writes to data stores; solid lines are inter-process data flows.

5. Entity–Relationship Diagram

Schema as defined in Laravel migrations under backend/database/migrations/. Relationships shown here are logical — the application links tables by session_id and by program slug rather than hard foreign-key constraints.

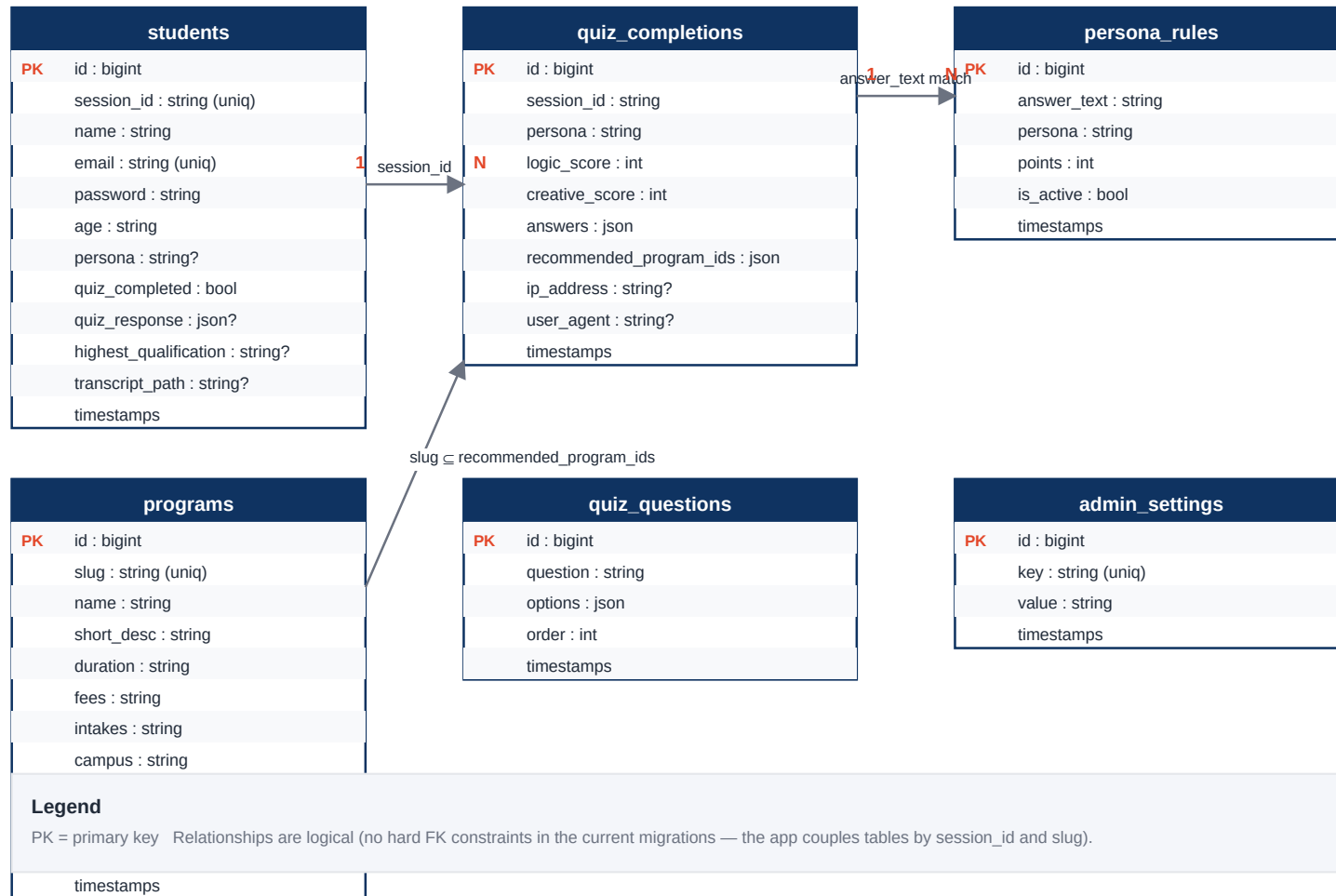
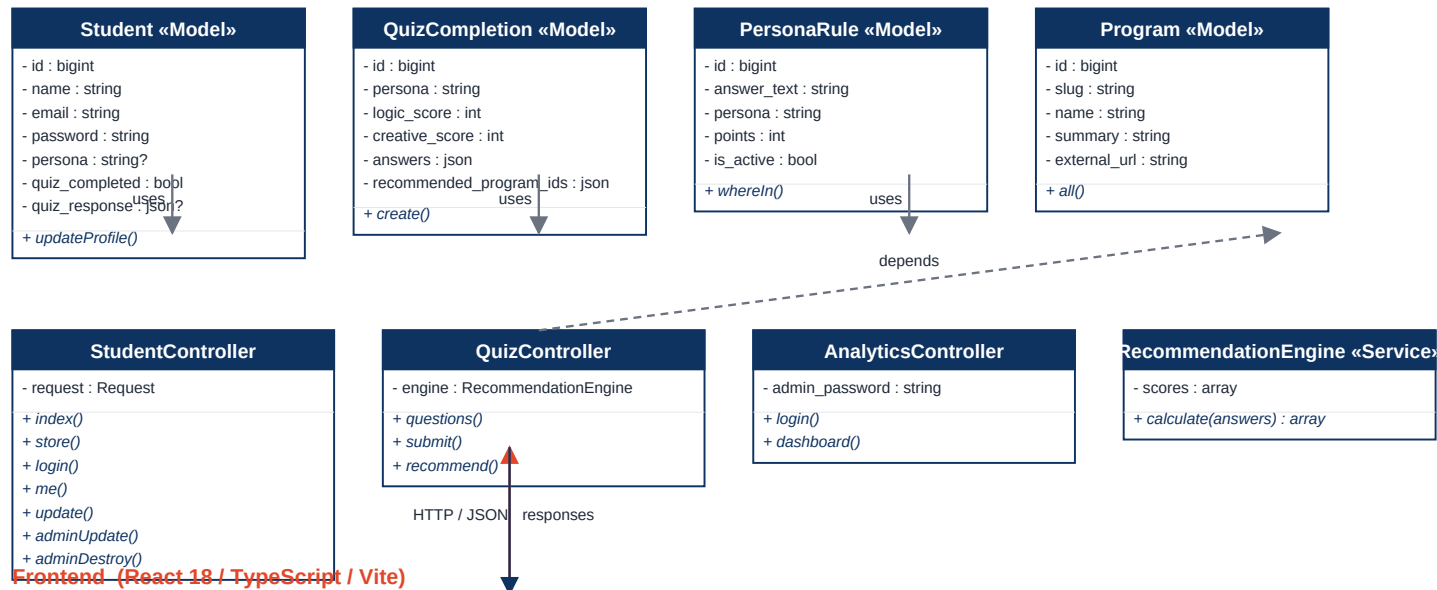


Figure 4 — ERD. Each box represents a MySQL table; PK marks the primary key. Bracketed cardinalities (1, N) show the multiplicity of the relationship.

6. UML Class Diagram

Layered view of the key classes / modules in both tiers. Top half is Laravel (Models, Controllers, Service). Bottom half is React (Context, API client, Engine, Pages). The two tiers communicate only over HTTP / JSON.

Backend (Laravel 11 / PHP 8.3)



Frontend (React 18 / TypeScript / Vite)

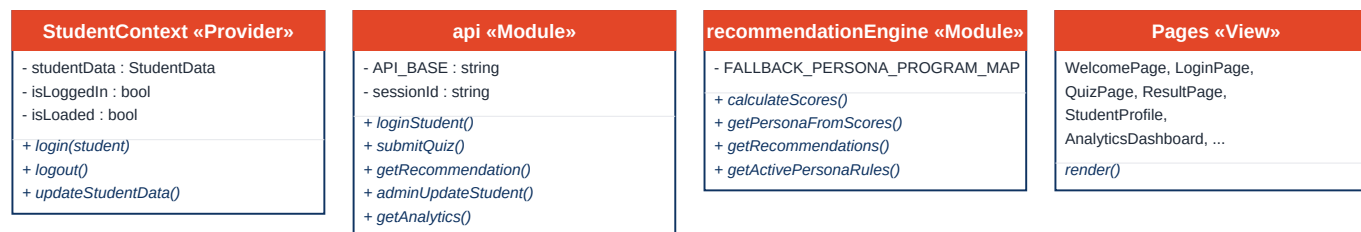


Figure 5 — UML Class Diagram. «Stereotype» tags clarify the role of each class (Model / Service / Provider / Module / View).